

Auditory setup

Parts and estimated cost

9 October 2026. Planned recordings in sedated sheep. Prices are in United States dollars; product names link to sellers or quote pages.

If we only need tones below 20 kilohertz, the basic setup is the least expensive starting point. If we need the paper’s full 1.4–32 kilohertz range, I suggest getting quotes for the compact setup on page 4. All four options use a compatible lab computer and our neural recorder. Labor and recorder-specific connections are extra.

1. Basic setup

Parts: \$346.79, or \$565.79 with a sound-level calibrator.

Computer → sound interface → powered speaker. The measurement microphone connects to the sound interface.

Item	Count	Seller / condition	Cost (\$)
Behringer UMC204HD sound interface	1	eBay / new	97.98
JBL 305P MkII powered speaker	1	Sweetwater / new	99.00
Dayton EMM-6 measurement microphone	1	eBay / new	66.93
Hosa HMIC-010 microphone cable	1	Sweetwater / new	17.99
Hosa HSS-005 audio cable, \$14.95 each	2	Sweetwater / new	29.90
Gator RI-MICTP-FBM microphone stand	1	Sweetwater / new	34.99
Subtotal without calibrator			346.79
Optional: REED R8090 sound-level calibrator	1	DigiKey / new	219.00
Subtotal with calibrator			565.79

- With a borrowed calibrator, allow \$426.79: parts plus an estimated \$80 for delivery and tax.
- Buying the calibrator brings the allowance to \$665.79, including an estimated \$100 reserve. The connection to our neural recorder is extra in both cases.

Timing

- We have not measured this setup’s timing error or drift. For scale, a 256-sample buffer at 48,000 samples per second lasts 5.3 milliseconds; that is a buffer example, not a measured interface delay.
- Measure sound arrival and each tone’s level at the ear. Sound travel adds about three milliseconds per metre, on top of speaker and converter delays.
- Record repeated timing markers with the neural data to measure accumulating mismatch. A microphone recorded through the sound interface still needs to be aligned with the neural recording.

For our discussion

- What frequency range do we need?
- What timing inputs does our recorder have?
- Which parts can we borrow, and what budget should we use?

The \$99 speaker price is a limited-time sale. Delivery/tax reserves are estimates. A sound output cannot be assumed compatible with a digital timing input. References: buffering; sound travel; delay estimates.

2. SONIC setup

Subtotal: \$7,357.58, plus the supplier quotes below.

This follows the paper’s USB-6361 sound device, SA1 amplifier and MF1 speaker for 1.4–32 kilohertz tones. It includes the paper’s satellite-based time reference, but uses our recorder in place of Connexus.

One sound output channel. Starred items are accessories added to make the chain usable; their exact part numbers are not identified in the paper. National Instruments makes the sound device; Tucker-Davis Technologies makes the speaker and amplifier.

Item	Count	Seller / condition	Cost (\$)
USB-6361 sound generation/recording, 782255-01	1	Newark / new	3,744.42
TimeMachines TM2500C time reference	1	New / backorder	799.99
MF1-M speaker kit	1	Supplier quote	Quote
SA1 speaker amplifier	1	Supplier quote	Quote
ZB1PS powered enclosure*	1	Supplier quote	Quote
781513-01 power supply*	1	Newark / new	123.42
780534-01 locking computer cable*	1	DigiKey / new	69.75
763830-01 United States power cord*	1	Newark / new	51.00
781233-02 cable noise filter*	1	DigiKey / new	38.25
Hosa BNC-58-103 output lead*	1	Pro Audio / new	13.53
Sound parts subtotal			4,840.36
Measurement kit (listed on page 4)	1	New amplifier	2,517.22
Subtotal, excluding quote items			7,357.58

Still needed

- Quotes for the MF1-M speaker, SA1 amplifier and ZB1PS powered enclosure. Confirm the speaker kit’s power and cable contents.
- Microphone and recorder connections, network wiring, delivery, tax and labor. The time reference is backordered.
- A seller-new 782255-01 on eBay for \$3,498.95 would save \$245.47; its accessories and actual condition need confirmation.

Timing

- SONIC reports 56 milliseconds for neural measurement, filtering and collecting data for the decoder. This excludes sensory processing and computer calculation time.
- The 56 milliseconds is not sound-to-neural timestamp error, and we cannot transfer that result to our recorder by buying the sound hardware.
- The paper does not quantify remaining alignment error or long-session drift. The time reference specifies below one microsecond plus network variation; that does not describe sound arrival at the ear.

The separate 11-millisecond result came from a one-session analysis. The sound device’s typical free-running clock scale at 25 degrees Celsius is 180 milliseconds per hour versus ideal time; actual drift relative to our recorder is unknown. SONIC recorded awake sheep. The effects of sedation on our responses and timing remain unmeasured.

References: SONIC version 1, pages 5, 7–9 and 11; sound device; time reference; speaker.

3. McGinley lab setup

Subtotal: \$10,068.97, plus the ES1 speaker quote.

The supplied slides use a desktop computer and equipment chassis connected to an ED1 driver and ES1 speaker. The list keeps both signal boards; our neural recorder replaces the Neuropixels equipment shown in the slides.

One sound output channel. Most chassis parts are used. The desktop needs a compatible free expansion slot. This route is worth considering if we already have substantial rack hardware.

Item	Count	Seller / condition	Cost (\$)
PXIe-1082 equipment chassis	1	eBay / used	2,650.00
PXIe-6363 signal input/output board	1	eBay / used	2,495.36
PXI-4461 audio module, hybrid 186900-11L	1	eBay / used	699.99
PXIe-8360 chassis connection module	1	eBay / used	493.20
PCIe-8361 computer connection card	1	eBay / listed new	197.00
779500-03 rack cable, 3 metres	1	eBay / listed new	89.85
SHC68-68-EPM cable, 192061C-02	1	eBay / used	89.99
BNC-2090A signal connection panel	1	eBay / used [‡]	613.77
ED1 driver + enclosure/power supply	1	eBay / used	209.06
ES1 speaker + matching cable	1	Supplier quote	Quote
Hosa BNC-58-103 output lead*	1	Pro Audio / new	13.53
Sound parts subtotal			7,551.75
Measurement kit (listed on page 4)	1	New amplifier	2,517.22
Subtotal, excluding quote items			10,068.97

Still needed

- An ES1 speaker quote with its matching high-voltage cable. The listed ED1 bundle includes its powered enclosure and supply.
- Microphone and recorder connections, any missing chassis power cord, delivery, tax and labor. Check the audio board's actual hybrid connector and calibration before buying.

Timing and frequency range

- The ES1 covers 4–110 kilohertz, so it misses the paper's 1.4–4 kilohertz tones. The audio board's flat output extends to 92.1 kilohertz at 204,800 samples per second.
- At that rate, the output filter adds a fixed 0.15625 milliseconds. Speaker and air delays are additional; this is not the system's timing accuracy.
- We have not measured alignment, drift or sound-level accuracy at the ear. The boards' clock settings, recorder connections and acoustic calibration need to be checked together.

The audio board's internal-clock scale is 72 milliseconds per hour versus ideal time, conditional on calibration and aging. A different reference changes this figure; drift relative to our recorder is unknown. The speaker's raw ± 11 -decibel response variation is not calibrated at-ear accuracy. The connection-panel seller lists no returns. The starred output lead is our addition.

References: supplied diagram; supplied parts slide; audio board, pages 1, 19 and 24; speaker; hybrid variant.

4. Compact setup

Subtotal: \$5,636.47, plus speaker, amplifier and enclosure quotes.

This is the option I suggest pricing if we need the full paper frequency range. It uses the computer-connected USB-6361, an SA1 amplifier and an MF2-M speaker, with a microphone near the ear and timing markers recorded with the neural data.

The main list uses a new microphone amplifier. Dagger-marked items make up the \$2,517.22 measurement kit also included on pages 2 and 3. Costs cover the counts listed.

Item	Count	Seller / condition	Cost (\$)
USB-6361 sound generation/recording, 781442-01	1	Newark / new	3,030.42
MF2-M speaker kit	1	Supplier quote	Quote
SA1 speaker amplifier	1	Supplier quote	Quote
ZB1PS powered enclosure	1	Supplier quote	Quote
Earthworks M50 measurement microphone [†]	1	eBay / new	1,299.00
Sound Devices MM-1 microphone amplifier [†]	1	B&H / new	945.00
REED R8090 sound-level calibrator [†]	1	DigiKey / new	219.00
Hosa HMIC-010 microphone cable [†]	1	Sweetwater / new	17.99
Gator RI-MICTP-FBM microphone stand [†]	1	Sweetwater / new	34.99
Panasonic LR6XWA/2SB battery, \$0.62 each [†]	2	DigiKey / new	1.24
Pomona EM4970-36# output lead	1	DigiKey / new	23.68
Hosa PHX-106F-BULK microphone connection lead	1	B&H / new	13.95
Yageo 100,000-ohm resistor, \$0.10 each	2	DigiKey / new	0.20
763830-01 United States power cord	1	Newark / new	51.00
Subtotal, excluding quote items			5,636.47

Still needed

- Quotes for the MF2-M speaker kit, SA1 amplifier and ZB1PS powered enclosure, plus the recorder connection, delivery, tax and labor. Device power/cable and speaker stand/cable are included; microphone wiring and bias returns need review.
- A used MM-1 at B&H for \$604.50 lowers the subtotal to \$5,295.97. It is listed with moderate wear and a 90-day seller warranty. The measurement kit would cost \$2,176.72.

Timing and calibration

- The speaker is nominally rated for 1–80 kilohertz and the microphone to 50 kilohertz. The calibrator's 0.5-decibel specification applies at one kilohertz only; it does not establish accuracy across all our tones.
- One possible test criterion is to align timestamps of sound arrival with the neural recording within one millisecond after correction. This has not been demonstrated and measures something different from SONIC's processing delay.
- Measure repeated markers, sound arrival and converter channel offsets. The setup has not been built or validated; its remaining alignment error and drift are unknown.

The sound device's clock scale is 180 milliseconds per hour versus ideal time, typical at 25 degrees Celsius; actual recorder-relative drift is unknown. Arduino or a field-programmable gate array would still need the analog sound hardware, acoustic calibration and program development.

References: speaker; microphone; microphone amplifier; calibrator; sound device; detailed comparison.